FILTER DESIGN TEST CASE

TC-SAW (Temperature Compensated SAW) - WiFi 6E (6 GHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0037
Revision:	Rev B.5
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Created Date:	2025-04-14
Last Modified:	2025-04-20
Status:	NEEDS REVIEW
Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a TC-SAW (Temperature Compensated SAW) filter targeting WiFi 6E (6 GHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 5462.7 MHz and bandwidth of 278.2 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: TC-SAW (Temperature Compensated SAW)
- Frequency Band: WiFi 6E (6 GHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: 128° YX LiNbO3
- Package: WLP (Wafer Level Package)
- Design Tool: CST Studio Suite 2024
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the TC-SAW (Temperature Compensated SAW) design targeting WiFi 6E (6 GHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 5462.7 MHz
Bandwidth (3 dB): 278.2 MHz
Insertion Loss Target: <= 3.1 dB
Return Loss Target: >= 16.0 dB
Out-of-Band Rejection: >= 22.0 dB
Isolation (Duplexer): >= 28.0 dB
Q Factor: 7644
Temperature Coefficient:-27.4 ppm/°C
Die Size: 2.6 x 1.3 mm
Wafer Size: 8-inch
Number of Resonators: 9
Electrode Material: Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the TC-SAW (Temperature Compensated SAW) template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 5462.7 MHz, BW 278.2 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.1 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 16388 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.1 dB
- Return loss in passband shall be >= 16.0 dB
- Out-of-band rejection at +/- 417 MHz offset >= 22.0 dB
- Group delay variation across passband shall be <= 5 ns
- Temperature drift over -40°C to +85°C shall be <= 18.7 MHz
- Power handling shall be >= +33 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 96%

6. TEST RESULTS

Measurement Date: 2025-04-20
Devices Tested: 32
Insertion Loss (meas): 2.84 dB
Return Loss (meas): 18.4 dB
Rejection (meas): 25.5 dB
Isolation (meas): 33.2 dB
Q Factor (meas): 7644
Group Delay Variation: 4.6 ns
Power Handling: +26 dBm
Die Yield: 98.4%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

a) The TC-SAW (Temperature Compensated SAW) design demonstrated below-target performance for WiFi 6E (6 GHz).
b) Insertion loss is marginally above target; electrode thickness optimization recommended.
c) Rejection at near-band offset requires additional resonator tuning.
d) Package parasitic effects were consistent with 3D EM model predictions.
e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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